INT421—Digital Signal Processing

Impulse Response

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Topics of Last Lecture

- Discrete-Time Systems
- System Characteristics
  1. Static (Memoryless) vs. Dynamic (with memory)
  2. Causal vs. Non-Causal
  3. BIBO Stable vs. BIBO unstable
  4. Linear vs. Non-linear
  5. Time-invariant vs. Time-variant

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2. Classification of LTI Systems
3. Convolution
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Impulse Response

- The impulse response $h[n]$ of a system $y[n] = T\{x[n]\}$ is defined as:

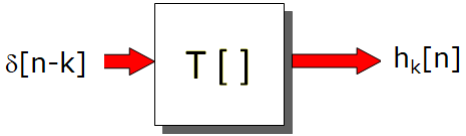
$$h[n] = T\{x[n]\}$$

$h[n]$ is the response (output) when the input is an impulse $\delta[n]$.

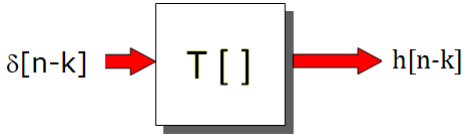


Impulse Response

- For an arbitrary system if the input is shifted by k, then the output is $h_k[n] = T\{\delta[n-k]\}$.

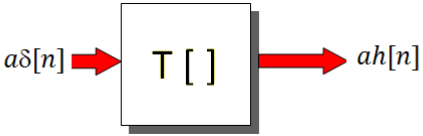


- If the system is time invariant, then a shift of k in the input will lead to the same shift in the output.



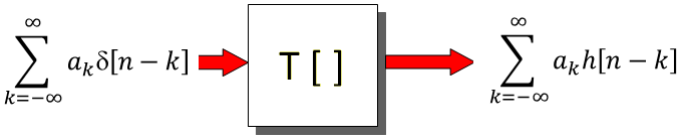
Impulse Response

- For a linear system if the input is scaled by α, then the output will be scaled by α.



- For a linear and time invariant system

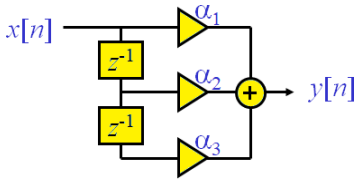
$$\alpha_1\delta[n-k_1]+\alpha_2\delta[n-k_2]=\alpha_1 h[n-k_1]+\alpha_2 h[n-k_2]$$

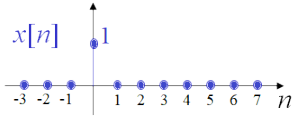


- **Linear Shift Invariant** system completely specified by $h[n]$

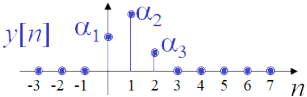
Impulse Response Example

Simple System





$x[n] = \delta[n]$ impulse



$y[n] = h[n]$ impulse response

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Classification of LTI Systems

- The causal discrete-time LTI systems classified into **two** broad categories:
  1. **non-recursive** systems are given by the following difference equation:

$$y[n] = \sum_{k=0}^{M} b_k x[n-k]$$

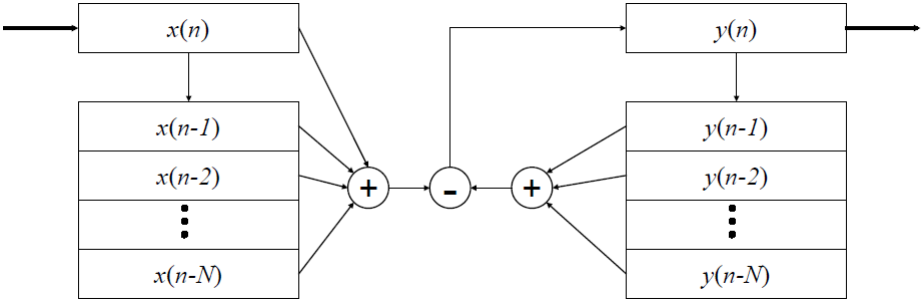
  2. **recursive** system refer to previous values of the output:

$$y[n] = -\sum_{l=1}^{N} a_l y[n-l] + \sum_{k=0}^{M} b_k x[n-k]$$

- The output of an LTI system represented by *linear constant-coefficient different equation* is found by using M input data observed up to now and N output data already found before. N is called the **ORDER** of the filter.

Classification of LTI Systems

- The Implementation achieved by two **first-in-first-out (FIFO)** memories (Queues).



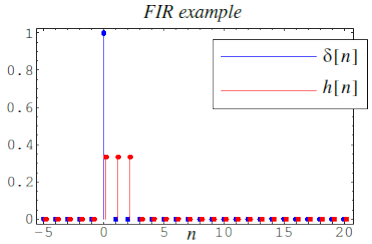
- By selecting properly the coefficients, we can get different kinds of digital filters (or digital signal processors).

Classification of LTI Systems

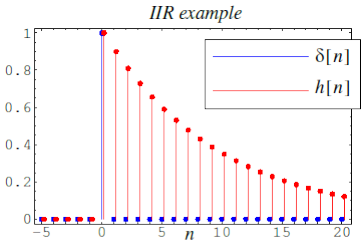
We will now label these two categories of LTI systems by their impulse response characteristics as:

1. Finite Impulse Response (**FIR**) systems,
2. Infinite Impulse Response (**IIR**) systems

.



$$y[n] = \frac{1}{3}x[n] + \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2]$$

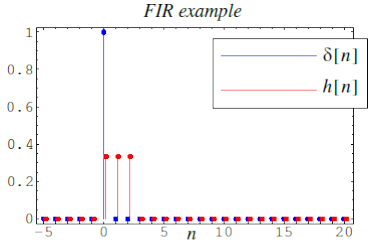


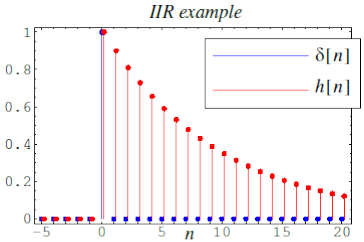
$$y[n] = (9/10)y[n-1] + x[n]$$

Classification of LTI Systems

1. A system is an FIR system if the outputs are calculated using the input data only.
   - If the coefficients a_k $(k = 1, 2 \ldots, N)$ are all zeros, the impulse response will be $h[k] = b_k$
2. A system is an IIR system if at least one a_k is not zero.

.





$$y[n] = \frac{1}{3}x[n] + \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2]$$

$$y[n] = (9/10)y[n-1] + x[n]$$

Quiz

Examples

Sketch the realizations of the following systems

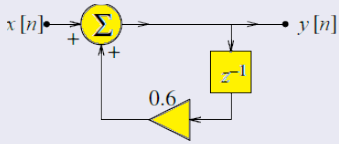
1. $y[n] - 0.6y[n-1] = x[n]$
2. $y[n] - \frac{1}{6}y[n-1] - \frac{1}{6}y[n-2] = 4x[n]$

Quiz

Solution

Sketch the realizations of the following systems

1. $y[n] - 0.6y[n-1] = x[n]$
2. $y[n] - \frac{1}{6}y[n-1] - \frac{1}{6}y[n-2] = 4x[n]$



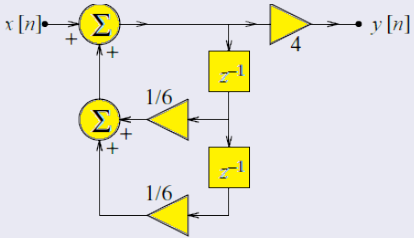


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Convolution

The LTI system is completely characterized by its **impulse response** as follows:

$$h[n] = T\{\delta[n]\}$$

$$y[n] = \sum_{k}^{\infty} x[k]h[n-k], \quad \textbf{convolution}$$

$$y[n] \triangleq x[n] * h[n]$$

Summation is symmetric in x and h, that if replace $m = n - k$ then

$$x[n] * h[n] = \sum_{m}^{\infty} x[n-m]h[m] = h[n] * x[n]$$

Convolution

Recall that: $x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$ then $T\{x[n]\} = T\{\sum_{k=-\infty}^{\infty} x[m]\delta[n-k]\}$

$$T\{x[n]\} = \sum_{k=-\infty}^{\infty} x[k]\, T\{\delta[n-k]\} \quad \textbf{Apply linearity}$$

$$T\{x[n]\} = \sum_{k=-\infty}^{\infty} x[k]h[k-n] \quad \textbf{Apply time-invariance}$$

$$y[n] = \sum_{k}^{\infty} x[k]h[n-k], \quad \textbf{Convolution}$$

The output of any *linear*, *time-invariant (LTI) systems* can be expressed as a **weighted sum of shifted impulse responses**

Convolution Sum Properties of LTI Systems

- **Commutative**:

$$x[n] * h[n] = h[n] * x[n]$$

- **Distributive**:

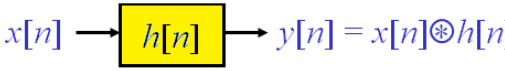
$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

- **Associative**:

$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$$

Interpreting Convolution

Passing a signal through a (LSI) system is equivalent to **convolving** it with the **system's impulse response**.



$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

$x[n] = \{2, 4, 6, 4, 2\}$ $\quad$ $h[n] = \{1, 2, -1, 3\}$

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2 Classification of LTI Systems

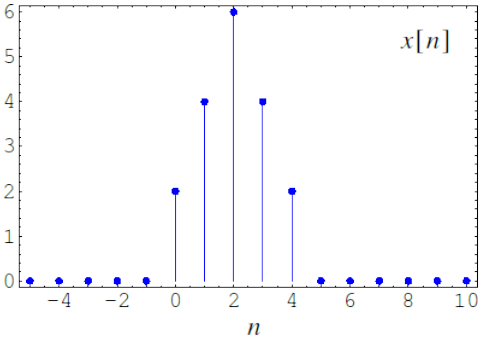
3 Convolution

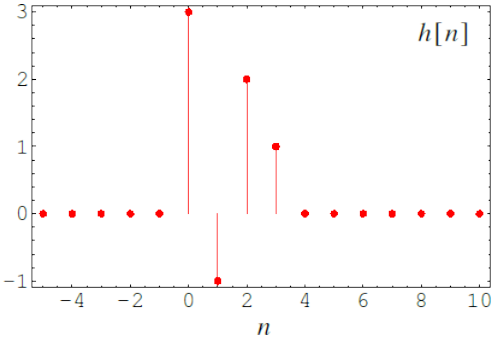
4 Computing the Convolution Sum

5 Summary

Computing the Convolution Sum

Two approaches to computing the convolution sum for the sample input sequence and finite impulse response function $h[n]$ plotted in next Figure.





First Approach

Applying $y[n] = \sum_k^{\infty} x[k]h[n-k]$ by time-reverse h, shift by n, take inner product against fixed signal x

n	k	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8
	$x[k]$	0	0	0	0	2	4	6	4	2	0	0	0	0
0	$h[0-k]$		1	2	-1	3	$y[0] = [0,0,0,2] \cdot [1,2,-1,3] = 6$							
1	$h[1-k]$			1	2	-1	3	$y[1] = [0,0,2,4] \cdot [1,2,-1,3] = 10$						
2	$h[2-k]$				1	2	-1	3	$y[2] = [0,2,4,6] \cdot [1,2,-1,3] = 18$					
3	$h[3-k]$					1	2	-1	3	$y[3] = 16$ [a]				
4	$h[4-k]$	$y[4] = 18$ [b]					1	2	-1	3				
5	$h[5-k]$	$y[5] = [6,4,2,0] \cdot [1,2,-1,3] = 12$						1	2	-1	3			
6	$h[6-k]$	$y[6] = [4,2,0,0] \cdot [1,2,-1,3] = 8$							1	2	-1	3		
7	$h[7-k]$	$y[7] = [2,0,0,0] \cdot [1,2,-1,3] = 2$								1	2	-1	3	
8	$h[8-k]$	$y[8] = [0,0,0,0] \cdot [1,2,-1,3] = 0$									1	2	-1	3

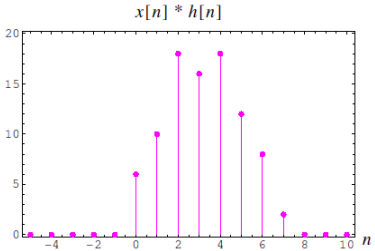
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First Approach

Convolution can be seen as a graphical process:

1. Plot $x[k]$ with dependent variable k
2. Plot $h[-k]$ with dependent variable k (h reflected around $k = 0$).
3. Plot $h[n-k]$ with dependent variable k (n can shift $h[n-k]$ from ∞ (all the way to the left) to ∞ (all the way to the right).
4. For each shift (i.e. n), compute $y[n] = \sum_k^{\infty} x[k]h[n-k]$ (i.e., multiply $x[k]h[n-k]$ and then sum the result).



Second Approach

Applying $y[n] = \sum_k^{\infty} h[k]x[n-k]$ by multiply each nonzero element of $h[k]$ by $x[n]$, and shift the resulting vectors to the right by k spaces, take inner product.

n	0	1	2	3	4	5	6	7	8
$x[n]$	2	4	6	4	2				
$h[n]$	3	-1	2	1					
$h[0]x[n-0]$	6	12	18	12	6				
$h[1]x[n-1]$		-2	-4	-6	-4	-2			
$h[2]x[n-2]$			4	8	12	8	4		
$h[3]x[n-3]$				2	4	6	4	2	
$y[n]$	6	10	18	16	18	12	8	2	0

Quiz-1

Examples

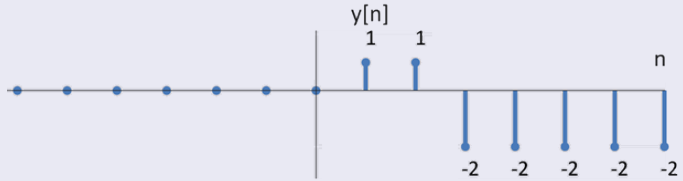
Consider the system defined by the input-output relationship

$$y[n] = \sum_{k=-\infty}^{m} x[k]$$

Compute the output for input $x[n] = \delta[n-1] - 3\delta[n-3]$

Quiz-1

Solution



Quiz-2

Examples

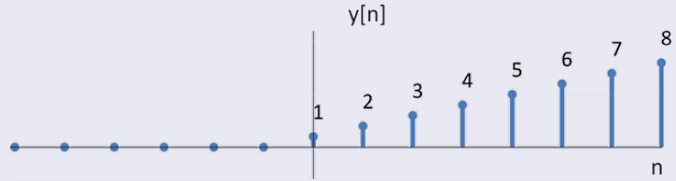
Consider the system defined by the input-output relationship

$$y[n] = \sum_{k=-\infty}^{m} x[k]$$

Compute the output for input $x[n] = u[n]$

Quiz-2

Solution



Quiz-3

Examples

Consider the system defined by the input-output relationship

$$y[n] = \frac{1}{2}(x[n] + x[n-1])$$

Compute the impulse response $h[n]$ of the system

Quiz-3

Solution

$$x[n] = \delta[n]$$

$$h[n] = \frac{1}{2}(\delta[n] + \delta[n-1])$$

Quiz-4

Examples

Consider the system defined by the input-output relationship

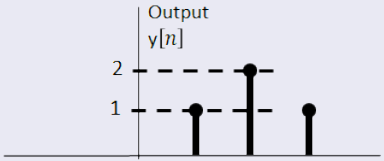
$$y[n] = \frac{1}{2}(x[n] + x[n-1])$$

Compute the convolution of:

$$h[n] = \frac{1}{2}(\delta[n] + \delta[n-1])$$
$$x[n] = 2\delta[n-1] + 2\delta[n-2]$$

Quiz-4

Solution



$$y[n] = \delta[n-1] + 2\,\delta[n-2] + \delta[n-3]$$

Quiz-5

Examples

Consider the system defined by the input-output relationship

$$y[n] = x[n] + \frac{1}{2}x[n-1], \quad \text{for all integer n.}$$

Let us find the response to

$$x[n] = \begin{cases} \frac{1}{3}, & \text{n=-1,} \\ 1, & \text{n=0,} \\ \frac{2}{3}, & \text{n=1,} \\ 0, & \text{otherwise} \end{cases}$$

Quiz-5

Solution

The impulse response $h[n]$ of the system is the response to the unit impulse

$$h[n] = \delta[n] + \frac{1}{2}x[n-1]$$

Therefore, the response to $x[n]$ is:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = x[-1]h[n+1] + x[0]h[n] + x[1]h[n-1], \text{ for all integer } n.$$

This leads to

$$y[n] = \begin{cases} \frac{1}{3}, & n\text{=-1,2,} \\ \frac{7}{6}, & n\text{=0,1,} \\ 0, & \text{otherwise} \end{cases}$$

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Summary

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- Systems represented by linear constant-coefficient difference equations.
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Any Questions?

